

GARDELEGEN, Germany

WMO: 103590

Lat: 52.5128N

Lon: 11.3942E

Elev: 48

StdP: 100.75

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.6	-10.9	-18.1	0.8	-13.7	-14.6	1.1	-9.6	13.1	6.2	11.5	5.8	0.6	30	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.8	19.0	27.8	18.1	26.0	17.3	19.8	27.2	18.9	25.9	17.9	24.6	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.4	22.6	16.0	11.5	22.1	15.1	10.8	21.1	56.9	27.3	53.5	26.0	50.6	24.6	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.7	DB	-16.9	33.0	7.0	2.3	-21.9	34.7	-26.0	36.1	-29.9	37.4	-35.0	39.0	(4)
(5)			WB	-17.5	22.0	6.8	1.0	-22.4	22.7	-26.4	23.3	-30.2	23.8	-35.1	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.4	-0.1	4.3	7.3	12.4	15.4	17.5	17.3	13.5	9.3	4.6	2.1	(6)
(7)		DBStd	7.37	5.97	5.47	3.90	3.66	3.67	3.45	3.20	3.07	2.68	3.71	3.71	4.15	(7)
(8)		HDD10.0	1347	298	284	180	96	18	1	0	0	2	57	166	246	(8)
(9)		HDD18.3	3583	556	517	436	330	186	102	55	58	146	281	413	503	(9)
(10)		CDD10.0	881	1	1	2	16	92	162	234	226	109	35	3	1	(10)
(11)		CDD18.3	74	0	0	0	0	2	13	31	26	2	1	0	0	(11)
(12)		CDH23.3	1086	0	0	0	1	69	223	418	339	31	4	0	0	(12)
(13)		CDH26.7	292	0	0	0	0	4	59	122	104	3	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.9	3.3	3.5	2.9	2.6	2.7	2.7	2.5	3.0	2.9	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	582	48	35	38	32	52	55	72	61	50	46	45	48	(15)
(16)		PrecMax	804	108	76	95	72	136	96	144	133	131	137	100	103	(16)
(17)		PrecMin	412	1	9	6	3	17	22	21	18	8	9	0	17	(17)
(18)		PrecStd	99	24	17	22	17	33	22	35	30	29	29	23	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	14.8	17.9	22.2	27.2	31.3	31.6	32.1	26.9	22.8	14.6	12.7	(19)
(20)			MCWB	9.9	10.5	11.0	13.8	16.1	18.7	19.7	20.1	17.3	17.2	11.0	9.8	(20)
(21)		2%	DB	9.8	11.3	14.2	19.4	25.4	28.2	29.8	29.5	23.6	19.1	12.5	11.0	(21)
(22)			MCWB	7.9	8.3	9.6	12.3	15.3	17.9	19.3	18.8	16.4	14.3	10.3	8.8	(22)
(23)		5%	DB	8.6	8.6	11.8	16.9	23.3	25.9	27.9	27.0	21.0	16.8	11.3	9.6	(23)
(24)			MCWB	6.8	6.4	7.9	11.0	14.6	16.8	18.5	18.0	15.0	13.0	9.4	8.0	(24)
(25)		10%	DB	7.2	6.6	10.1	14.6	21.1	23.6	25.8	24.7	19.1	15.1	10.1	8.2	(25)
(26)			MCWB	5.5	4.5	7.2	9.7	13.7	15.8	17.8	17.3	14.3	12.0	8.5	6.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.7	11.6	14.5	17.7	20.6	21.4	21.2	18.5	17.3	11.6	10.0	(27)
(28)			MCDWB	11.8	14.1	16.1	21.0	24.5	26.4	29.6	30.0	24.8	21.8	13.1	12.1	(28)
(29)		2%	WB	8.3	8.2	10.0	12.9	16.3	18.6	20.0	19.6	17.0	15.1	10.7	9.1	(29)
(30)			MCDWB	9.7	10.9	13.5	18.2	23.4	26.6	27.5	27.1	22.4	18.9	12.3	10.7	(30)
(31)		5%	WB	6.9	6.4	8.7	11.4	15.1	17.4	19.0	18.7	15.8	13.4	9.7	8.0	(31)
(32)			MCDWB	8.5	8.5	11.2	16.2	21.6	24.5	26.2	25.2	19.9	16.3	11.3	9.5	(32)
(33)		10%	WB	5.6	4.7	7.3	10.1	14.0	16.2	17.9	17.8	14.8	12.3	8.5	6.7	(33)
(34)			MCDWB	7.2	6.4	9.8	14.1	20.3	22.8	24.9	23.7	18.5	14.8	10.0	8.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	7.3	7.8	10.4	11.9	11.0	11.6	11.5	9.5	8.4	5.9	4.8	(35)
(36)			MCDBR	5.7	9.3	11.8	14.7	16.3	16.1	16.4	16.4	13.8	11.7	7.6	6.2	(36)
(37)			MCWBR	4.8	7.2	8.2	8.8	8.5	7.9	8.1	8.2	8.3	8.0	6.2	5.1	(37)
(38)		5% WB	MCDWR	5.2	8.9	10.9	13.2	14.5	14.1	14.6	13.7	11.4	10.8	7.2	5.8	(38)
(39)			MCWBR	4.6	7.1	7.9	8.2	8.1	7.5	7.7	7.4	7.2	7.8	6.1	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.340	0.376	0.413	0.413	0.410	0.429	0.422	0.386	0.360	0.337	0.311	(40)	
(41)		taud	2.354	2.339	2.254	2.205	2.251	2.279	2.239	2.272	2.355	2.406	2.394	2.352	(41)	
(42)		Ebn at Noon	670	765	803	818	839	844	818	802	788	732	640	606	(42)	
(43)		Edh at Noon	61	82	108	128	128	126	129	119	97	76	59	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.65	1.36	2.43	3.99	4.92	5.30	4.98	4.21	3.10	1.75	0.79	0.50	(44)	
(45)		RadStd	0.07	0.19	0.23	0.47	0.56	0.42	0.69	0.34	0.35	0.18	0.08	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.38	N/A	N/A	N/A	N/A	+0.53	N/A	-158	N/A	N/A

Nomenclature: See separate page